

Roshan Razak

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SKILLS

Core Programming: Python (async, type hinting, Pydantic), Java, JavaScript, TypeScript, SQL

AI & Orchestration: LangChain, LangGraph, RAG Pipelines, Agentic Workflows, Prompt Engineering, LLM Evaluation, Ollama

Vector Databases: pgvector (PostgreSQL), ChromaDB; document chunking, embedding models, semantic search

Backend & Web: FastAPI, Spring Boot, ReactJS, Laravel, Django, REST APIs

Databases: PostgreSQL, MySQL, Redis, DynamoDB

DevOps & Infrastructure: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD for AI applications, AWS (Lambda, S3, EC2, Bedrock)

EXPERIENCE

Cawosh Ltd, United Kingdom

Software Engineer, August 2024 – Present

- Developed and maintained Python backend services on AWS (Lambda, API Gateway), designing asynchronous data processing pipelines and integrating AI-ready endpoints for production inference workloads
- Built and optimised ETL pipelines feeding structured and unstructured data into downstream ML workflows, implementing document parsing and data normalisation stages for AI feature consumption
- Identified and resolved a bottleneck in a serverless data pipeline, cutting monthly infrastructure costs by 45% while maintaining processing throughput
- Implemented rigorous unit and integration testing across sprint cycles using Python test frameworks, improving deployment confidence and ensuring reliability of AI-serving infrastructure
- Automated secrets rotation via AWS IAM and enforced strict type-hinting and Pydantic validation patterns across backend modules, maintaining audit readiness for financial compliance

Capgemini Engineering, India

Associate Consultant, October 2018 – February 2023

- Built and maintained Java / Spring Boot microservices for high-volume financial transaction processing, resolving a critical sequencing bug that caused data drift across distributed services
- Re-architected long-running PostgreSQL queries for a global banking client, improving data retrieval speeds by 40% through strategic indexing and query optimisation
- Established testing suites and coding standards adopted team-wide, reducing production-level bugs by 25% within 18 months; awarded STAR Performer recognition

PROJECTS

Agentic AI Underwriting Pipeline - LangGraph / RAG / FastAPI

- Architected a multi-step agentic workflow using LangGraph to orchestrate autonomous underwriting decisions, with branching logic for risk assessment, document analysis, and human-in-the-loop review gates
- Built a production RAG pipeline with document chunking strategies, embedding model selection, and retrieval optimisation against a policy corpus, achieving 95% accuracy on semantic policy lookups
- Integrated ChromaDB as the vector store for semantic search, with FastAPI serving the inference API and Pydantic models enforcing structured output validation across all agent responses
- Implemented evaluation and testing harnesses to measure retrieval precision, answer accuracy, and latency across varying document corpus sizes

MemoryMesh - Multi-Tenant Conversational AI Platform

- Designed and built a multi-tenant conversational AI application with a three-layer memory architecture: Redis for session cache, pgvector for semantic retrieval via embedding similarity search, and Claude Haiku for long-term memory compression

- Developed the backend with FastAPI and LangChain, using asynchronous Python endpoints with Pydantic type-hinted request/response models, and PostgreSQL with pgvector for vector database operations
- Containerised the full stack with Docker Compose for one-command deployment, exposed via Cloudflare Tunnels; built a React MemoryDebugPanel to visually surface memory layer hits during live demos

Homelab Infrastructure as Code - Proxmox / Terraform / Docker

- Architected a Proxmox-based homelab managed via Terraform, running containerised services including Jenkins CI/CD, Ollama for local LLM inference, and automated deployment pipelines
- Achieved zero-open-port remote access using Cloudflare Tunnels with Nginx Proxy Manager, SSL termination, and MFA/SSO integration across all services

EDUCATION

Staffordshire University, July 2024

MSc Computer Science (Software Engineering)

- Designed and evaluated a Hybrid Reinforcement Learning system for personalised recommendations, integrating Collaborative Filtering with Content-Based Filtering to improve accuracy over either approach alone

Kannur University, July 2017

B.Tech in Computer Science and Engineering

- Engineered a Real-Time Incremental Sentiment Analysis system (Project RISTS) to process high-velocity social media streams using short-text classification algorithms